CLASSIFICATION & DECISIONING / PRODUCTION MLOPS

From churn scores to repeatable retention decisions

Deliver cost-aware retention recommendations through an API, monitor quality and keep model changes controlled.

WHY MOVE BEYOND THE NOTEBOOK?

01

Reliable delivery

Make churn scores and action recommendations available to downstream applications.

02

Less manual work

Automate repeatable data checks, training and release steps, with traceable results.

03

Controlled change

Monitor quality, evaluate new candidates and recover when release verification fails.

PROOF FROM A WORKING REFERENCE SYSTEM

Quality tracked

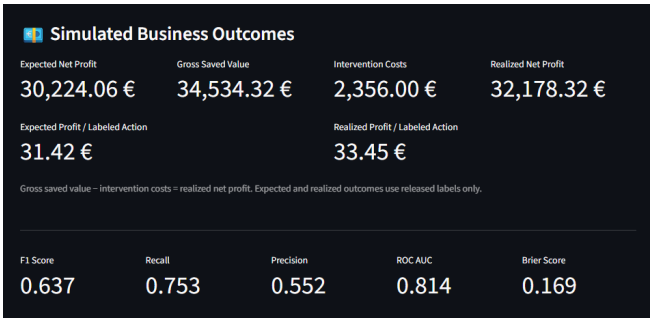
Post-shift mean rolling F1: 0.569 static versus 0.585 adaptive.

1 of 3 approved

Three retraining runs led to one Champion promotion.

Rollback verified

Previous API revision: 100% traffic and HTTP 200 readiness.



Business-aware monitoring

Local model quality and simulated retention outcomes shown together.



Verified application rollback

Cloud Run traffic and readiness; follow-up prediction probe checked lineage.

WHAT WAS BUILT

FastAPI decision service, Prefect pipelines and governed promotion. Private MLflow on Cloud Run, Cloud SQL metadata, GCS artifacts, Terraform and keyless GitHub Actions. Local model and business monitoring.

EVIDENCE SCOPE

Independent reference project using public Telco data and audited synthetic target drift. Business outcomes assume intervention uplift; additional simulated value was approximately €17. Cloud deployment and rollback successfully verified.

Have a working model, but no reliable deployment?

Start with a Notebook-to-MLOps Assessment: a code review and prioritized delivery roadmap.

Steffen Lauterbach | Freelance MLOps Engineer

GitHub: mlops-churn-prediction

LinkedIn / Contact